Inverse Statistical Geometry

Constructibility, Non-Identifiability, and the Epistemics of Relational Data

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Contents

1 Introduction

Classical statistics is primarily forward-facing.

The traditional paradigm is:

$$\text{data} \to \text{statistics.}$$

Examples include:

- datasets producing covariance matrices,
- vectors producing correlations,
- embeddings producing cosine similarities,
- time series producing autocorrelation structures.

This paper studies the inverse direction:

$$\text{target relational structure} \to \text{constructible data.}$$

Algorithms A–D established explicit constructions for vectors possessing prescribed:

- cosine similarities,
- Pearson correlations,
- centered angular relations,
- precomputed correlation geometries.

The existence of such algorithms suggests a broader mathematical principle:

Relational statistical structures are constructible.

This paper investigates the consequences of that principle across:

- statistics,
- data science,
- machine learning,
- economics,
- finance,
- political science,
- warfare economics,
- international relations,
- philosophy of science,
- epistemology.

The central claim is not that all datasets are synthetic.
Rather, the claim is substantially more precise:

> Observed relational statistical structure alone may be insufficient to uniquely identify the latent generating mechanism of a dataset.

This is the central thesis of inverse statistical geometry.
The paper ends with "The End".

2 Relational Geometry

2.1 Cosine Similarity

Given vectors

$$u, v \in \mathbb{R}^n,$$

cosine similarity is

$$\mathrm{cs}(u, v) = \frac{u^\top v}{|u||v|}.$$

Algorithms A and B demonstrated explicit constructions of vectors satisfying:

$$\mathrm{cs}(u, v) = s,$$

for arbitrary target similarity

$$s \in [-1, 1].$$

2.2 Pearson Correlation

Given vectors

$$u, v \in \mathbb{R}^n,$$

Pearson correlation is

$$\rho(u, v) = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2}\sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}.$$

Algorithms C and D established explicit constructions satisfying:

$$\rho(u, v) = c,$$

for arbitrary target correlation

$$c \in [-1, 1].$$

2.3 Centered Geometry

Define the centering matrix

$$P = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top.$$

Then:

$$u_c = Pu, \qquad v_c = Pv.$$

Pearson correlation becomes:

$$\rho(u, v) = \frac{u_c^\top v_c}{|u_c||v_c|}.$$

Thus correlation equals cosine similarity inside the centered hyperplane

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

3 The Constructibility Principle

Definition 3.1 (Constructibility Principle)**.** *A relational structure is constructible if one can explicitly generate realizations satisfying prescribed relational constraints.*

Theorem 3.2 (Constructibility of Relational Constraints)**.** *For every admissible target cosine similarity*

$$s \in [-1, 1]$$

and every admissible target Pearson correlation

$$c \in [-1, 1],$$

there exist explicit constructive algorithms that generate vectors satisfying those constraints.

Proof. Algorithms A–D provide explicit constructions for vectors satisfying arbitrary target relational constraints.

Therefore these relational structures are algorithmically synthesizable.

□

The significance of this result is not merely computational.
It changes the epistemic status of relational statistics themselves.

4 Inverse Statistical Geometry

4.1 Forward Statistics

Traditional statistics assumes:

$$D \to S(D),$$

where:

- D denotes a dataset,
- $S(D)$ denotes statistical summaries.

Examples include:

$$D \to \Sigma,$$

where

$$\Sigma$$

is a covariance matrix.

4.2 Inverse Statistics

Inverse statistical geometry studies the reverse mapping:

$$S \to D.$$

The central question becomes:

> Given a target statistical structure, can one explicitly construct datasets realizing it?

Algorithms A–D answer this affirmatively for important classes of relational statistics.

5 Non-Uniqueness and Manifold Structure

Theorem 5.1 (Infinite Solution Families)**.** *For fixed:*

$$u \in \mathbb{R}^n,$$

and target correlation:

$$c \in (-1, 1),$$

there exist infinitely many vectors

$$v$$

satisfying:

$$\rho(u, v) = c.$$

Proof. The solution set lies on a centered angular shell:

$$S^{n-3} \subset H.$$

For fixed u and c,
the admissible centered vectors form

$$c\hat{u} + \sqrt{1 - c^2} S^{n-3}.$$

This set is homeomorphic to S^{n-3}.
For $n \geq 3$, the sphere contains infinitely many points.
Therefore infinitely many solutions exist.

□

This implies:

relational constraints underdetermine realization.

This is a geometric consequence rather than a philosophical assertion.

6 Underdetermination of Generators

Theorem 6.1 (Underdetermination of Generators)**.** *Let*

$$S$$

denote a relational statistic.
Suppose there exist distinct datasets

$$D_1 \neq D_2$$

such that

$$S(D_1) = S(D_2).$$

Then observation of

$$S(D)$$

alone is insufficient to uniquely identify the generating dataset.

Proof. Suppose

$$S(D_1) = S(D_2) = s,$$

with

$$D_1 \neq D_2.$$

Then observation of

$$s$$

cannot distinguish between

$$D_1$$

and

$$D_2.$$

Therefore the generating realization is not uniquely identified by the observed relational statistic alone.

□

Theorem 6.2 (Underdetermination of Generators)**.** *Let*

$$S$$

denote a relational statistic.
If there exist infinitely many datasets

$$D$$

satisfying

$$S(D) = s,$$

then observation of

$$s$$

alone is insufficient to identify a unique generating dataset.

Proof. Suppose two distinct datasets

$$D_1 \neq D_2$$

satisfy

$$S(D_1) = S(D_2) = s.$$

Then observation of

$$s$$

cannot distinguish between them.
Therefore the generating realization is not uniquely identified.

□

7 Observational Equivalence

The underdetermination theorem implies the possibility of observational equivalence between distinct latent generators.

Suppose two generative systems

$$\mathcal{G}_1, \qquad \mathcal{G}_2,$$

produce datasets

$$D_1, \qquad D_2,$$

satisfying:

$$S(D_1) = S(D_2).$$

Then:

$$\mathcal{L}(S(D_1)) = \mathcal{L}(S(D_2)).$$

where $\mathcal{L}$ denotes the induced law of the observable statistics.

Consequently:

observed relational structure alone may not uniquely determine the latent generator.

This becomes especially severe when:

- covariance structures are constructible,
- correlations are constructible,
- embedding geometries are constructible,
- synthetic datasets preserve statistical realism.

8 Implications for Data Science

8.1 Synthetic Statistical Realism

If covariance and correlation structures are constructible, then statistically realistic datasets become constructible.

This affects:

- benchmark datasets,
- stress-testing systems,
- adversarial machine learning,
- simulation frameworks,
- generative AI systems.

8.2 Dataset Non-Identifiability

Traditional data science often assumes:

$$\text{statistics} \rightarrow \text{latent process}.$$

Inverse statistical geometry weakens this assumption.

Distinct generators may produce:

- identical correlations,

- identical covariance structures,
- identical low-order moments,
- similar geometric relations.

Thus:

$$\text{statistics} \not\Rightarrow \text{unique generator}.$$

9 Machine Learning Implications

Modern machine learning already exhibits geometric dependence:

- embedding manifolds,
- cosine similarity neighborhoods,
- latent semantic geometry,
- attention-space structure.

Inverse statistical geometry implies:

latent geometries themselves may be synthesizable.

This has implications for:

- foundation models,
- representation learning,
- synthetic pretraining corpora,
- adversarial embeddings.

10 Economic and Financial Implications

10.1 Synthetic Financial Structures

Modern finance relies heavily upon:

- covariance matrices,
- factor models,
- correlation stress testing,
- Monte Carlo simulation.

If dependence structures are explicitly constructible, then:

- synthetic market scenarios become easier to generate,
- adversarial market simulations become feasible,
- stress-testing scenario generation becomes more tractable.

10.2 Systemic Risk

Suppose institutions rely upon

$$\Sigma$$

as a sufficient risk descriptor.

If many distinct latent systems share similar covariance structures, then

$$\Sigma$$

may fail to identify systemic fragility uniquely.

This introduces:

- hidden systemic equivalence classes,
- latent instability manifolds,
- adversarial covariance mimicry.

11 Political and Geopolitical Implications

Modern geopolitical systems increasingly rely upon:

- intelligence fusion,
- predictive analytics,
- statistical surveillance,
- AI-assisted forecasting.

If statistical realism is constructible, then:

- synthetic geopolitical signals become feasible,
- adversarial informational structures become cheaper,
- observational ambiguity increases.

This affects:

- information warfare,
- cyber conflict,
- strategic deception,
- synthetic intelligence operations.

12 Warfare Economics

Military systems increasingly depend upon:

- probabilistic targeting,
- statistical reconnaissance,
- autonomous battlefield inference,
- AI-generated strategic simulation.

Inverse statistical geometry implies:

statistically plausible observational signatures may be constructible.

Examples include:

- synthetic logistical patterns,
- adversarial sensor correlations,
- statistically plausible decoy networks.

Thus warfare increasingly becomes:

a competition over relational statistical control.

13 Epistemological Consequences

The strongest implication is epistemic rather than technological.
Classical empiricism assumes:

1. reality exists,
2. observations sample reality,
3. datasets accumulate truth.

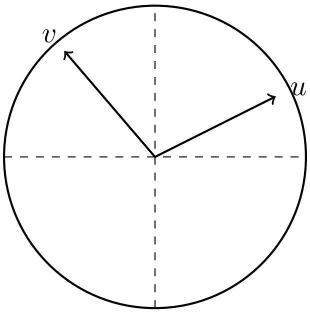
Inverse statistical geometry suggests an alternative structure:

1. relational constraints define admissible manifolds,
2. many datasets satisfy those manifolds,
3. observations may underdetermine generators.

This does not invalidate empirical science.
However, it weakens the assumption that:

$$\text{observation} \to \text{unique latent mechanism.}$$

14 Geometric Visualization



Centered angular shell inside the mean-zero hyperplane

The figure illustrates the geometric structure underlying inverse correlation synthesis.

15 Toward a General Theory of Inverse Statistical Geometry

Algorithms A–D likely represent only the beginning of a broader theory.

Potential extensions include:

- inverse covariance synthesis,
- inverse mutual-information synthesis,
- inverse copula construction,
- inverse graphical-model synthesis,
- inverse latent-factor geometry,
- inverse manifold construction.

The broader principle is:

> relational statistics are not merely descriptive summaries; they are constructive geometric objects.

16 Limitations

The existence of inverse constructions does not imply:

- all datasets are synthetic,
- all generators are observationally equivalent,
- higher-order statistics are constructible,
- complete identification is impossible.

The results establish only that certain relational structures admit multiple realizations.

17 Conclusion

Algorithms A–D demonstrate that important relational statistical structures — including cosine similarity and Pearson correlation — are explicitly constructible through inverse geometric methods.

Consequently:

> observed relational structure alone may not uniquely determine the latent generating mechanism of a dataset.

This shifts relational statistics from:

- descriptive observables,

  to

- constructive geometric entities.

The resulting framework — inverse statistical geometry — has implications across statistics, machine learning, economics, finance, warfare economics, political science, philosophy, and epistemology.

Tables

Algorithmic Progression

Algorithm	Relational Structure	Geometry	Computational Style
A	Cosine Similarity	Euclidean Sphere	Stochastic
B	Cosine Similarity	Euclidean Sphere	Precomputed Basis
C	Pearson Correlation	Centered Hyperplane	Stochastic
D	Pearson Correlation	Centered Hyperplane	Precomputed Basis

Epistemic Consequences

Domain	Consequence
Data Science	Dataset Non-Identifiability
Machine Learning	Constructible Embedding Geometry
Finance	Synthetic Covariance Systems
Geopolitics	Adversarial Statistical Realism
Warfare	Statistical Camouflage
Philosophy	Underdetermined Generators

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Glossary

Constructibility
The ability to explicitly generate realizations satisfying prescribed relational constraints.

Correlation Shell
The manifold of vectors satisfying a fixed target correlation relative to an anchor vector.

Inverse Statistical Geometry
The study of constructing datasets from prescribed statistical structures.

Observational Equivalence
The phenomenon whereby distinct latent generators produce statistically similar observables.

Relational Constraint
A prescribed statistical or geometric relation between objects.

Statistical Realism
The preservation of important statistical structures within synthetic datasets.

The End